

DDA3020 Machine Learning

Tutorial: Optimization

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Instructions

This tutorial covers the core concepts of Optimization in Machine Learning, including convex sets, convex functions, convex optimization problems, unconstrained minimization (gradient descent), and constrained minimization (Lagrangian duality, KKT conditions). Show your working clearly for all calculation and derivation questions.

Useful Formulas

- First-order condition for convexity: $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in \text{dom } f$
- Second-order condition for convexity: $\nabla^2 f(\mathbf{x}) \succeq 0$ (Positive Semi-Definite) for all $\mathbf{x} \in \text{dom } f$
- Gradient Descent Update Rule: $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - t^{(k)} \nabla f(\mathbf{x}^{(k)})$, where $t^{(k)}$ is the step size
- Lagrangian Function: $L(\mathbf{x}, \mathbf{u}, \mathbf{v}) = f(\mathbf{x}) + \sum_{i=1}^m u_i h_i(\mathbf{x}) + \sum_{j=1}^r v_j \ell_j(\mathbf{x})$
- KKT Conditions:
 - Stationarity: $0 \in \partial f(\mathbf{x}) + \sum_{i=1}^m u_i \partial h_i(\mathbf{x}) + \sum_{j=1}^r v_j \partial \ell_j(\mathbf{x})$
 - Complementary slackness: $u_i \cdot h_i(\mathbf{x}) = 0$ for all i
 - Primal feasibility: $h_i(\mathbf{x}) \leq 0, \ell_j(\mathbf{x}) = 0$ for all i, j
 - Dual feasibility: $u_i \geq 0$ for all i

Exercises

Multiple Choice Questions

Q1. Which of the following sets is NOT a convex set?

- (A) A line segment between any two points
- (B) The solution set of linear equations $\{\mathbf{x} \mid A\mathbf{x} = \mathbf{b}\}$
- (C) A kidney-shaped region in 2D space
- (D) A convex polygon

Q2. Which of the following functions is strictly convex on $\mathbb{R}$?

- (A) $f(x) = 2x + 3$
- (B) $f(x) = e^x$
- (C) $f(x) = \log x$
- (D) $f(x) = |x|$

Q3. For a twice continuously differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, what is the necessary and sufficient condition for it to be convex?

- (A) Its gradient $\nabla f(\mathbf{x})$ is zero everywhere
- (B) Its Hessian matrix $\nabla^2 f(\mathbf{x})$ is positive semi-definite for all $\mathbf{x}$
- (C) Its Hessian matrix $\nabla^2 f(\mathbf{x})$ is negative semi-definite for all $\mathbf{x}$
- (D) Its domain is an open set

Q4. In a standard convex optimization problem, which of the following statements is true?

- (A) The objective function must be concave
- (B) The inequality constraint functions must be affine
- (C) The equality constraint functions must be affine
- (D) Any locally optimal point is not necessarily globally optimal

Q5. Which of the following is NOT a KKT condition for constrained optimization?

- (A) Stationarity
- (B) Complementary slackness
- (C) Positive definiteness of the Hessian
- (D) Primal and dual feasibility

Calculation Questions

Q6. Consider the quadratic function $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top P\mathbf{x} + \mathbf{q}^\top \mathbf{x} + r$, where $P = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$ and

$$\mathbf{q} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

- (1) Calculate the gradient $\nabla f(\mathbf{x})$ and the Hessian $\nabla^2 f(\mathbf{x})$.
 - (2) Determine whether $f(\mathbf{x})$ is a convex function. Justify your answer.
 - (3) Find the global minimum point $\mathbf{x}^*$ of $f(\mathbf{x})$.
- Q7. Consider the unconstrained optimization problem $\min f(x_1, x_2) = x_1^2 + 2x_2^2$. We apply the gradient descent method starting from $\mathbf{x}^{(0)} = (2, 1)^\top$.
- (1) Calculate the gradient $\nabla f(\mathbf{x}^{(0)})$.
 - (2) If we use a fixed step size $t = 0.1$, calculate the next point $\mathbf{x}^{(1)}$.

Short Answer Questions

- Q8. Briefly explain the difference between exact line search and backtracking line search in the gradient descent method. What are the advantages and disadvantages of each?
- Q9. In machine learning, why do we prefer convex optimization problems over non-convex ones? Give two examples of machine learning models that result in convex optimization problems.

Derivation Questions

Q10. Consider the constrained optimization problem:

$$\begin{aligned} \min_{x_1, x_2} \quad & x_1^2 + x_2^2 \\ \text{subject to} \quad & x_1 + x_2 - 2 \geq 0 \end{aligned}$$

- (1) Rewrite the problem in the standard minimization form (with ≤ 0 inequality constraints).
- (2) Write down the Lagrangian function $L(x_1, x_2, u)$.
- (3) Derive the KKT conditions for this problem.
- (4) Solve the KKT conditions to find the optimal solution (x_1^*, x_2^*) and the optimal dual variable u^* .

Solutions

Multiple Choice Questions

- Q1. **C.** A kidney-shaped region is non-convex because a line segment connecting two points in the "lobes" of the kidney will pass outside the region.
- Q2. **B.** $f(x) = e^x$ has $f''(x) = e^x > 0$ for all x , so it is strictly convex. $f(x) = 2x + 3$ is affine (convex but not strictly). $f(x) = \log x$ is concave. $f(x) = |x|$ is convex but not strictly convex at $x = 0$.
- Q3. **B.** The second-order condition states that a twice differentiable function is convex if and only if its Hessian is positive semi-definite ($\nabla^2 f(\mathbf{x}) \succeq 0$).
- Q4. **C.** In a standard convex optimization problem, the objective function must be convex, inequality constraints must be convex, and equality constraints must be affine. Also, any locally optimal point is globally optimal.
- Q5. **C.** The KKT conditions consist of stationarity, complementary slackness, primal feasibility, and dual feasibility. Positive definiteness of the Hessian is a condition for strict convexity, not a KKT condition.

Calculation Questions

Q6. (1) $\nabla f(\mathbf{x}) = P\mathbf{x} + \mathbf{q} = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4x_1 + 2x_2 - 1 \\ 2x_1 + 2x_2 + 1 \end{bmatrix}.$

$$\nabla^2 f(\mathbf{x}) = P = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}.$$

- (2) To check if $f(\mathbf{x})$ is convex, we check if P is positive semi-definite. The principal minors are $\Delta_1 = 4 > 0$ and $\Delta_2 = \det(P) = 4(2) - 2(2) = 8 - 4 = 4 > 0$. Since all principal minors are positive, P is positive definite, so $f(\mathbf{x})$ is strictly convex.

- (3) Since $f(\mathbf{x})$ is convex, the global minimum is found where $\nabla f(\mathbf{x}) = 0$.

$$P\mathbf{x} + \mathbf{q} = 0 \implies P\mathbf{x} = -\mathbf{q} \implies \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Solving the system: $4x_1 + 2x_2 = 1$ and $2x_1 + 2x_2 = -1$.

Subtracting the second from the first: $2x_1 = 2 \implies x_1 = 1$.

Substitute $x_1 = 1$ into the second: $2(1) + 2x_2 = -1 \implies 2x_2 = -3 \implies x_2 = -1.5$.

So $\mathbf{x}^* = (1, -1.5)^\top$.

Q7. (1) $\nabla f(\mathbf{x}) = \begin{bmatrix} 2x_1 \\ 4x_2 \end{bmatrix}$. At $\mathbf{x}^{(0)} = (2, 1)^\top$, $\nabla f(\mathbf{x}^{(0)}) = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$.

(2) $\mathbf{x}^{(1)} = \mathbf{x}^{(0)} - t\nabla f(\mathbf{x}^{(0)}) = \begin{bmatrix} 2 \\ 1 \end{bmatrix} - 0.1 \begin{bmatrix} 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 - 0.4 \\ 1 - 0.4 \end{bmatrix} = \begin{bmatrix} 1.6 \\ 0.6 \end{bmatrix}$.

Short Answer Questions

Q8. **Exact line search** finds the optimal step size t that exactly minimizes the objective function along the search direction: $t = \arg \min_{t>0} f(\mathbf{x} + t\Delta\mathbf{x})$. It provides the maximum possible decrease in one step, but solving this 1D optimization problem can be computationally expensive.

Backtracking line search is an inexact method that starts with a large step size (e.g., $t = 1$) and iteratively reduces it by a factor $\beta \in (0, 1)$ until the sufficient decrease condition (Armijo condition) is met: $f(\mathbf{x} + t\Delta\mathbf{x}) < f(\mathbf{x}) + \alpha t \nabla f(\mathbf{x})^\top \Delta\mathbf{x}$. It is computationally cheaper per iteration and guarantees convergence, making it more practical for complex functions.

Q9. We prefer convex optimization problems because any locally optimal point is guaranteed to be a globally optimal point. This means optimization algorithms like gradient descent will not get stuck in suboptimal local minima and will reliably find the global best solution.

Examples of machine learning models that result in convex optimization problems include Linear Regression (with MSE loss) and Logistic Regression (with Cross-Entropy loss).

Derivation Questions

Q10. (1) Standard form requires $h_i(\mathbf{x}) \leq 0$. The constraint $x_1 + x_2 - 2 \geq 0$ becomes $2 - x_1 - x_2 \leq 0$.

So, $\min f(\mathbf{x}) = x_1^2 + x_2^2$ subject to $h_1(\mathbf{x}) = 2 - x_1 - x_2 \leq 0$.

(2) The Lagrangian function is $L(x_1, x_2, u) = f(\mathbf{x}) + uh_1(\mathbf{x}) = x_1^2 + x_2^2 + u(2 - x_1 - x_2)$.

(3) The KKT conditions are:

- Stationarity: $\nabla_{\mathbf{x}} L = 0 \implies \frac{\partial L}{\partial x_1} = 2x_1 - u = 0$ and $\frac{\partial L}{\partial x_2} = 2x_2 - u = 0$.
- Complementary slackness: $u(2 - x_1 - x_2) = 0$.
- Primal feasibility: $2 - x_1 - x_2 \leq 0 \implies x_1 + x_2 \geq 2$.

- Dual feasibility: $u \geq 0$.

(4) From stationarity, we have $x_1 = u/2$ and $x_2 = u/2$.

Case 1: $u = 0$. Then $x_1 = 0, x_2 = 0$. But $x_1 + x_2 = 0 \not\geq 2$, which violates primal feasibility.

Case 2: $u > 0$. By complementary slackness, $2 - x_1 - x_2 = 0 \implies x_1 + x_2 = 2$.

Substitute $x_1 = u/2$ and $x_2 = u/2$ into the constraint: $u/2 + u/2 = 2 \implies u = 2$.

Since $u = 2 > 0$, dual feasibility is satisfied.

Then $x_1^* = 2/2 = 1$ and $x_2^* = 2/2 = 1$.

The optimal solution is $(x_1^*, x_2^*) = (1, 1)$ with optimal dual variable $u^* = 2$.